MATH120: Applications of Integration

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Learning Objectives

- Find the area between two curves using integration
- Compute the average value of a function over an interval
- Set up and evaluate volumes of revolution (disk and washer methods)

Simplify each expression completely. Show all steps and circle your final answer.

Area under a curve

1. Find the area of the region bounded by f(x) = x^2, x = 1, x = 2, and the x-axis.

$$\int_1^2 x^2\,dx$$

Answer: ____

2. Find the area of the region bounded by f(x) = x^3, x = 1, x = 3, and the x-axis.

$$\int_1^3 x^3\,dx$$

Answer: ____

3. Find the area of the region bounded by f(x) = x^2, x = 1, x = 4, and the x-axis.

$$\int_1^4 x^2\,dx$$

Answer: ____

4. Find the area of the region bounded by f(x) = x^2, x = 0, x = 2, and the x-axis.

$$\int_0^2 x^2\,dx$$

Answer: ____

5. Find the area of the region bounded by f(x) = x^3, x = 1, x = 3, and the x-axis.

$$\int_1^3 x^3\,dx$$

Answer: ____

6. Find the area of the region bounded by f(x) = x^3, x = 0, x = 2, and the x-axis.

$$\int_{0}^{2} x^3 \, dx$$

Answer: ____

7. Find the area of the region bounded by f(x) = x^3, x = 0, x = 3, and the x-axis.

$$\int_{0}^{3} x^3 \, dx$$

Answer: ____

8. Find the area of the region bounded by f(x) = x^3, x = 1, x = 3, and the x-axis.

$$\int_{1}^{3} x^3 \, dx$$

Answer: ____

9. Find the area of the region bounded by f(x) = x^3, x = 1, x = 2, and the x-axis.

$$\int_{1}^{2} x^3 \, dx$$

Answer: ____

10. Find the area of the region bounded by f(x) = x^2, x = 0, x = 2, and the x-axis.

$$\int_{0}^{2} x^2 \, dx$$

Answer: ____

Total accumulated change

11. The rate of sales is r(t) = 2t^2 units/day. Find the total number of units sold from day 0 to day 6.

$$\int_{0}^{6} x^2 \, dx$$

Answer: ____

12. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 0 to 6.

$$\int_{0}^{6} x^1 \, dx$$

Answer: ____

13. The rate of sales is r(t) = 2t^2 units/day. Find the total number of units sold from day 0 to day 4.

$$\int_0^4 x^2 \, dx$$

Answer: ____

14. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 1 to 6.

$$\int_1^6 x^1 \, dx$$

Answer: ____

15. The rate of sales is r(t) = 2t^2 units/day. Find the total number of units sold from day 1 to day 3.

$$\int_1^3 x^2 \, dx$$

Answer: ____

16. A factory's output rate is O(t) = x^2 (hundreds of units/hour). Find the total output over hours 1 to 4.

$$\int_1^4 x^2 \, dx$$

Answer: ____

17. The rate of sales is r(t) = 4t^1 units/day. Find the total number of units sold from day 1 to day 5.

$$\int_1^5 x^1 \, dx$$

Answer: ____

18. A factory's output rate is O(t) = x^2 (hundreds of units/hour). Find the total output over hours 2 to 6.

$$\int_2^6 x^2 \, dx$$

Answer: ____

19. The rate of sales is r(t) = 2t^1 units/day. Find the total number of units sold from day 0 to day 5.

$$\int_0^5 x^1 \, dx$$

Answer: ____

20. A factory's output rate is O(t) = x^2 (hundreds of units/hour). Find the total output over hours 2 to 5.

$$\int_2^5 x^2 \, dx$$

Answer:

21. The rate of sales is r(t) = 6t^1 units/day. Find the total number of units sold from day 0 to day 4.

$$\int_0^4 x^1\,dx$$

Answer: ______________________________

22. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 2 to 6.

$$\int_2^6 x^1\,dx$$

Answer: ______________________________

23. The rate of sales is r(t) = 5t^1 units/day. Find the total number of units sold from day 0 to day 5.

$$\int_0^5 x^1\,dx$$

Answer: ______________________________

24. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 0 to 6.

$$\int_0^6 x^1\,dx$$

Answer: ______________________________

25. The rate of sales is r(t) = 5t^1 units/day. Find the total number of units sold from day 1 to day 3.

$$\int_1^3 x^1\,dx$$

Answer: ______________________________

26. A factory's output rate is O(t) = x^2 (hundreds of units/hour). Find the total output over hours 1 to 4.

$$\int_1^4 x^2\,dx$$

Answer: ______________________________

27. The rate of sales is r(t) = 2t^1 units/day. Find the total number of units sold from day 1 to day 6.

$$\int_1^6 x^1\,dx$$

Answer: ______________________________

28. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 2 to 4.

$$\int_2^4 x^1\,dx$$

Answer:

29. The rate of sales is r(t) = 2t^2 units/day. Find the total number of units sold from day 1 to day 6.

$$\int_1^6 x^2\,dx$$

Answer: ____

30. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 1 to 4.

$$\int_1^4 x^1\,dx$$

Answer: ____

Topics: Total accumulated change, Area under a curve. All answers verified by independent computation.

Solutions

Area under a curve

1. Find the area of the region bounded by f(x) = x^2, x = 1, x = 2, and the x-axis.

$$\int_1^2 x^2\,dx$$

→ Area = integral from 1 to 2 of x^2 dx.
→ = [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.

Answer: $\frac{8-1}{3} = 7/3$

2. Find the area of the region bounded by f(x) = x^3, x = 1, x = 3, and the x-axis.

$$\int_1^3 x^3\,dx$$

→ Area = integral from 1 to 3 of x^3 dx.
→ = [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.

Answer: $\frac{81-1}{4} = 20$

3. Find the area of the region bounded by f(x) = x^2, x = 1, x = 4, and the x-axis.

$$\int_1^4 x^2\,dx$$

→ Area = integral from 1 to 4 of x^2 dx.
→ = [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.

Answer: $\frac{64-1}{3} = 21$

4. Find the area of the region bounded by f(x) = x^2, x = 0, x = 2, and the x-axis.

$$\int_0^2 x^2\,dx$$

→ Area = integral from 0 to 2 of x^2 dx.
→ = [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.

Answer: $\frac{8-0}{3} = 8/3$

5. Find the area of the region bounded by f(x) = x^3, x = 1, x = 3, and the x-axis.

$$\int_1^3 x^3\,dx$$

→ Area = integral from 1 to 3 of x^3 dx.
→ = [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.

Answer: $\frac{81-1}{4} = 20$

6. Find the area of the region bounded by f(x) = x^3, x = 0, x = 2, and the x-axis.

$$\int_0^2 x^3\,dx$$

→ *Area = integral from 0 to 2 of x^3 dx.*

→ *= [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.*

Answer: $\dfrac{16-0}{4} = 4$

7. Find the area of the region bounded by f(x) = x^3, x = 0, x = 3, and the x-axis.

$$\int_0^3 x^3\,dx$$

→ *Area = integral from 0 to 3 of x^3 dx.*

→ *= [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.*

Answer: $\dfrac{81-0}{4} = 81/4$

8. Find the area of the region bounded by f(x) = x^3, x = 1, x = 3, and the x-axis.

$$\int_1^3 x^3\,dx$$

→ *Area = integral from 1 to 3 of x^3 dx.*

→ *= [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.*

Answer: $\dfrac{81-1}{4} = 20$

9. Find the area of the region bounded by f(x) = x^3, x = 1, x = 2, and the x-axis.

$$\int_1^2 x^3\,dx$$

→ *Area = integral from 1 to 2 of x^3 dx.*

→ *= [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.*

Answer: $\dfrac{16-1}{4} = 15/4$

10. Find the area of the region bounded by f(x) = x^2, x = 0, x = 2, and the x-axis.

$$\int_0^2 x^2\,dx$$

→ *Area = integral from 0 to 2 of x^2 dx.*

→ *= [x^{n+1}/({n+1})] from {lo} to {hi} = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} sq units.*

Answer: $\dfrac{8-0}{3} = 8/3$

Total accumulated change

11. The rate of sales is r(t) = 2t^2 units/day. Find the total number of units sold from day 0 to day 6.

$$\int_0^6 x^2\,dx$$

→ Total units = integral from 0 to 6 of 2t^2 dt.
*→ = {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ Total = 72 units (Note: multiply by 2 if a > 1).

Answer: $\frac{216-0}{3} = 72$

12. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 0 to 6.

$$\int_0^6 x^1\,dx$$

→ Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.
→ = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.

Answer: $\frac{36-0}{2} = 18$

13. The rate of sales is r(t) = 2t^2 units/day. Find the total number of units sold from day 0 to day 4.

$$\int_0^4 x^2\,dx$$

→ Total units = integral from 0 to 4 of 2t^2 dt.
*→ = {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ Total = 64/3 units (Note: multiply by 2 if a > 1).

Answer: $\frac{64-0}{3} = 64/3$

14. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 1 to 6.

$$\int_1^6 x^1\,dx$$

→ Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.
→ = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.

Answer: $\frac{36-1}{2} = 35/2$

15. The rate of sales is r(t) = 2t^2 units/day. Find the total number of units sold from day 1 to day 3.

$$\int_1^3 x^2\,dx$$

→ Total units = integral from 1 to 3 of 2t^2 dt.
*→ = {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ Total = 26/3 units (Note: multiply by 2 if a > 1).

Answer: $\frac{27-1}{3} = 26/3$

16. A factory's output rate is O(t) = x^2 (hundreds of units/hour). Find the total output over hours 1 to 4.

$$\int_1^4 x^2\,dx$$

→ Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.
→ = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.

Answer: $\frac{64-1}{3} = 21$

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17. The rate of sales is r(t) = 4t^1 units/day. Find the total number of units sold from day 1 to day 5.

$$\int_1^5 x^1\,dx$$

→ Total units = integral from 1 to 5 of 4t^1 dt.
*→ = {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ Total = 12 units (Note: multiply by 4 if a > 1).

Answer: $\frac{25-1}{2} = 12$

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18. A factory's output rate is O(t) = x^2 (hundreds of units/hour). Find the total output over hours 2 to 6.

$$\int_2^6 x^2\,dx$$

→ Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.
→ = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.

Answer: $\frac{216-8}{3} = 208/3$

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19. The rate of sales is r(t) = 2t^1 units/day. Find the total number of units sold from day 0 to day 5.

$$\int_0^5 x^1\,dx$$

→ Total units = integral from 0 to 5 of 2t^1 dt.
*→ = {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ Total = 25/2 units (Note: multiply by 2 if a > 1).

Answer: $\frac{25-0}{2} = 25/2$

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20. A factory's output rate is O(t) = x^2 (hundreds of units/hour). Find the total output over hours 2 to 5.

$$\int_2^5 x^2\,dx$$

→ Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.
→ = ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.

Answer: $\frac{125-8}{3} = 39$

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21. The rate of sales is r(t) = 6t^1 units/day. Find the total number of units sold from day 0 to day 4.

$$\int_0^4 x^1\, dx$$

→ *Total units = integral from 0 to 4 of 6t^1 dt.*
→ *= {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ *Total = 8 units (Note: multiply by 6 if a > 1).*

Answer: $\frac{16-0}{2} = 8$

22. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 2 to 6.

$$\int_2^6 x^1\, dx$$

→ *Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.*
→ *= ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.*

Answer: $\frac{36-4}{2} = 16$

23. The rate of sales is r(t) = 5t^1 units/day. Find the total number of units sold from day 0 to day 5.

$$\int_0^5 x^1\, dx$$

→ *Total units = integral from 0 to 5 of 5t^1 dt.*
→ *= {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ *Total = 25/2 units (Note: multiply by 5 if a > 1).*

Answer: $\frac{25-0}{2} = 25/2$

24. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 0 to 6.

$$\int_0^6 x^1\, dx$$

→ *Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.*
→ *= ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.*

Answer: $\frac{36-0}{2} = 18$

25. The rate of sales is r(t) = 5t^1 units/day. Find the total number of units sold from day 1 to day 3.

$$\int_1^3 x^1\, dx$$

→ *Total units = integral from 1 to 3 of 5t^1 dt.*
→ *= {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ *Total = 4 units (Note: multiply by 5 if a > 1).*

Answer: $\frac{9-1}{2} = 4$

26. A factory's output rate is O(t) = x^2 (hundreds of units/hour). Find the total output over hours 1 to 4.

$$\int_1^4 x^2\,dx$$

→ *Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.*
→ *= ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.*

Answer: $\frac{64-1}{3} = 21$

27. The rate of sales is r(t) = 2t^1 units/day. Find the total number of units sold from day 1 to day 6.

$$\int_1^6 x^1\,dx$$

→ *Total units = integral from 1 to 6 of 2t^1 dt.*
→ *= {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ *Total = 35/2 units (Note: multiply by 2 if a > 1).*

Answer: $\frac{36-1}{2} = 35/2$

28. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 2 to 4.

$$\int_2^4 x^1\,dx$$

→ *Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.*
→ *= ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.*

Answer: $\frac{16-4}{2} = 6$

29. The rate of sales is r(t) = 2t^2 units/day. Find the total number of units sold from day 1 to day 6.

$$\int_1^6 x^2\,dx$$

→ *Total units = integral from 1 to 6 of 2t^2 dt.*
→ *= {a} * [t^{n+1}/({n+1})] from {lo} to {hi} = {a} * ({hi_pow} - {lo_pow}) / ({n+1}).*
→ *Total = 215/3 units (Note: multiply by 2 if a > 1).*

Answer: $\frac{216-1}{3} = 215/3$

30. A factory's output rate is O(t) = x^1 (hundreds of units/hour). Find the total output over hours 1 to 4.

$$\int_1^4 x^1\,dx$$

→ *Total = integral from {lo} to {hi} of x^{n} dx = [x^{n+1}/({n+1})] from {lo} to {hi}.*
→ *= ({hi_pow} - {lo_pow}) / ({n+1}) = {answer_defint210} hundred units.*

Answer: $\frac{16-1}{2} = 15/2$